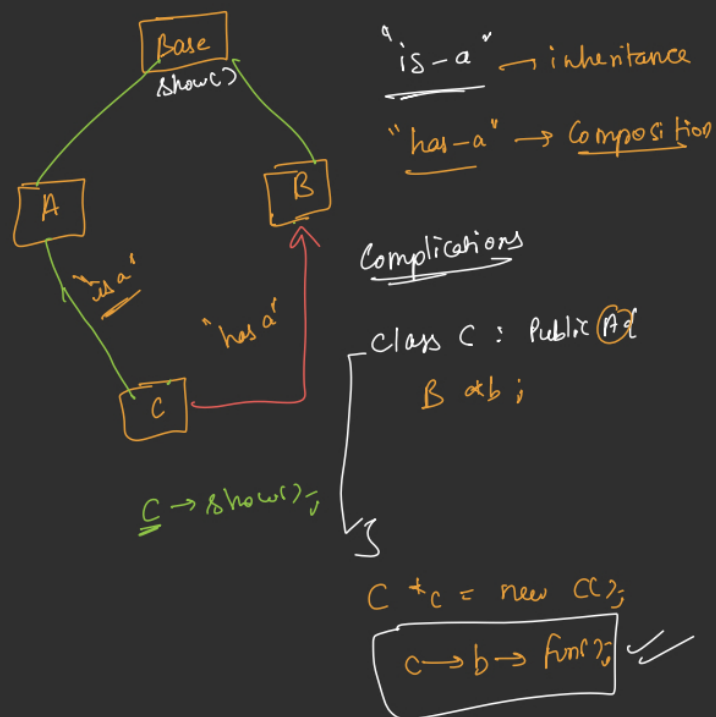
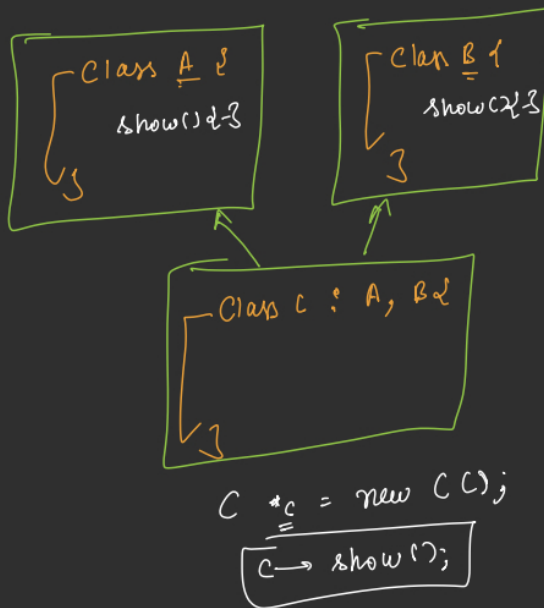
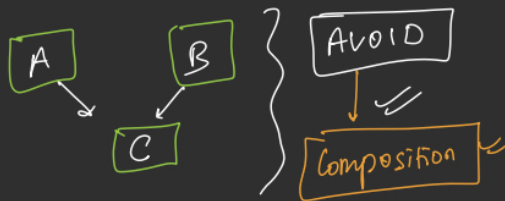
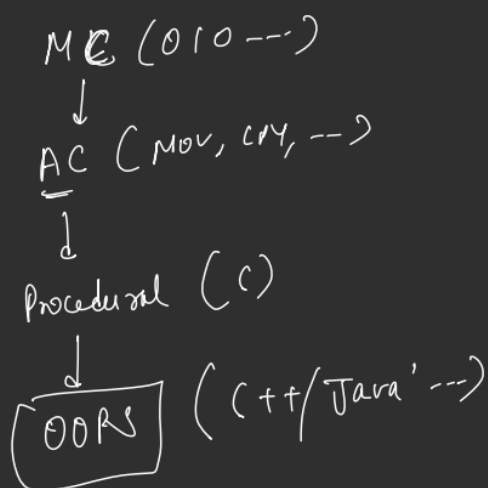
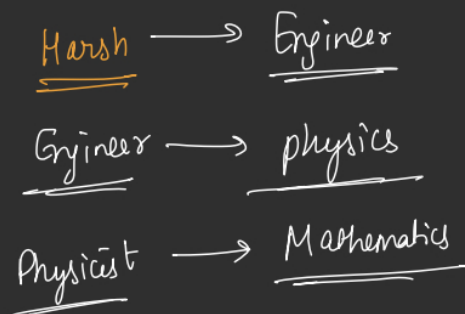
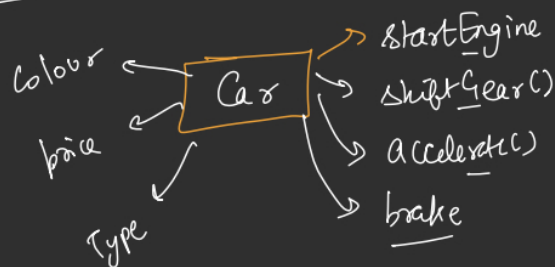
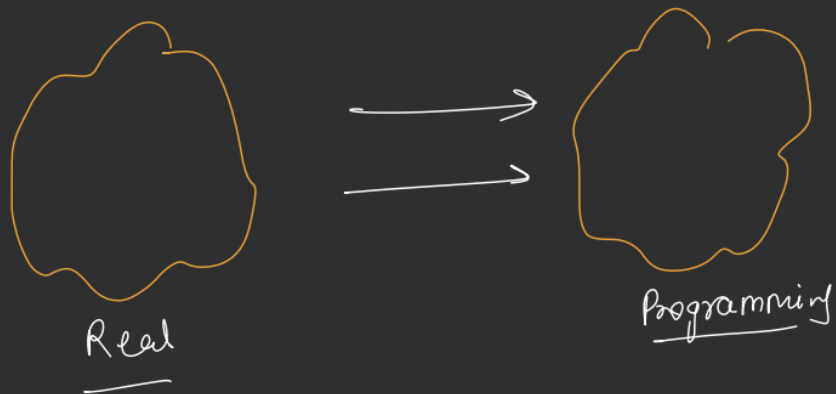


Diamond Problem

Multiple Inheritance



Abstraction

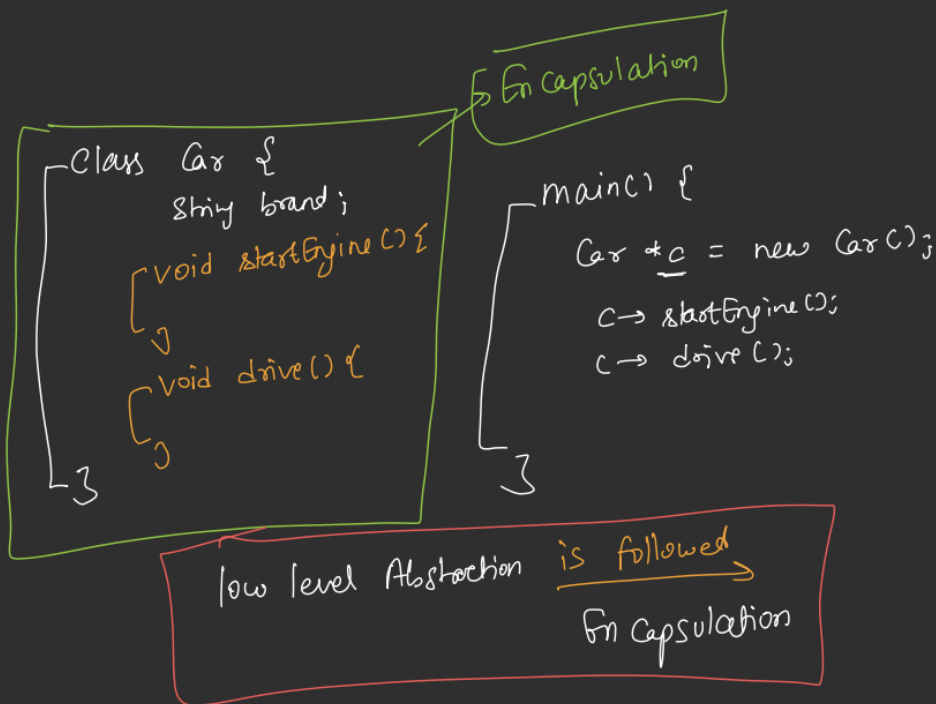


Abstraction

- ↳ Hiding unnecessary Details
 - ↳ Hiding implementation details
- from someone who just want to use it..

C++ / any OOPS language.

- ↳ low-level Abstraction $\Rightarrow$ Hiding implementation
- ↳ High-level Abstraction $\Rightarrow$ separating "WHAT" from "How"



High level Abstraction

↓
It separates the "WHAT" from "How".

```

class Car {
    void startEngine() {
    }
    void drive() {
    }
}

```

How

```

main() {
    Car *c = new Car();
    c -> startEngine();
    c -> drive();
}

```

"WHAT"

```

class Car {
    virtual void drive() = 0;
}

```

Abstract
virtual method

```

class PetrolCar : public Car {
    void drive() { }
}

```

How

```

class ElectricCar : public Car {
    void drive() { }
}

```

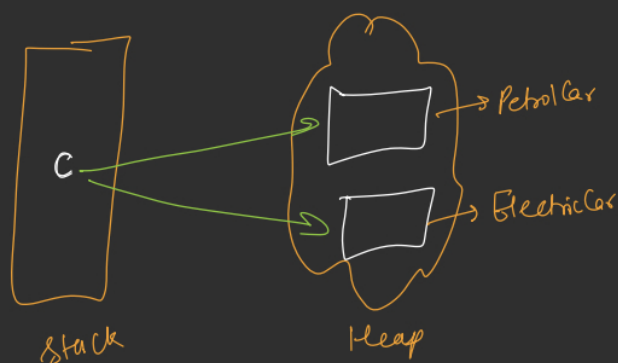
How

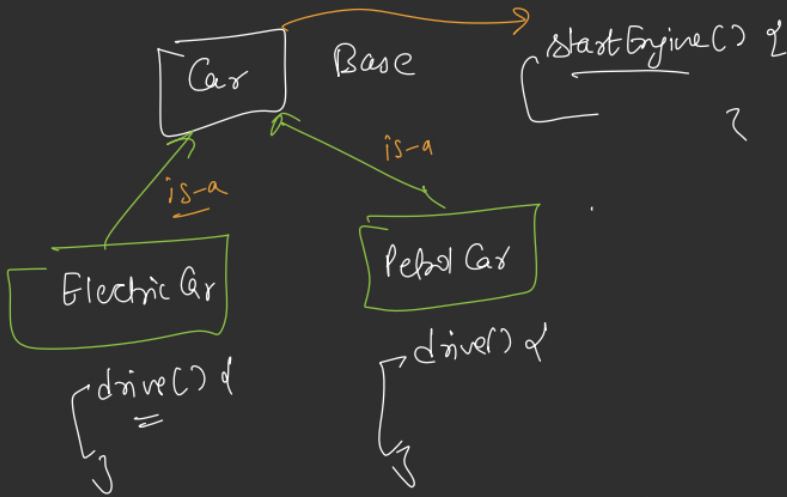
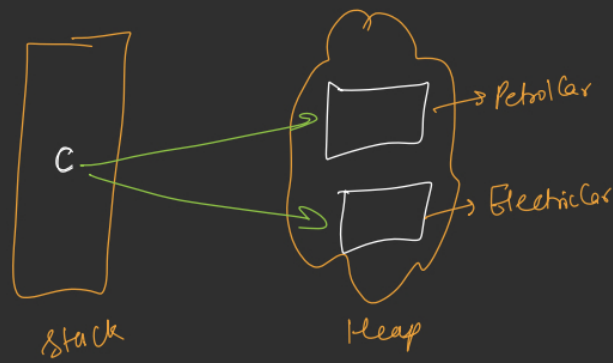
```

main() {
    PetrolCar();
    Car *c = new ElectricCar();
    c -> drive();
}

```

"WHAT"





main()

```

Car *c = new Car();
Car *c2 = new PetrolCar();
  
```

Abstraction

Hiding

unnecessary/
implementation
details

Encapsulation

Security